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# Adaptive Multi-Modal Anomaly Detection for 5G/B5G Networks Using Correlation Graphs, Dynamic Thresholding, and Explainability-Driven Drift Detection

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• **ABSTRACT** Fifth-generation (5G) and beyond-5G (B5G) cellular networks face increasingly sophisticated distributed denial-of-service (DDoS) attacks that expose critical limitations in existing machine-learning (ML)-based intrusion detection systems. Current approaches, including recent gradient-boosted and deep-learning systems, treat network performance metrics as independent signals, rely on a single static decision threshold, and employ explainability tools only for offline auditing. This paper presents an adaptive multi-modal anomaly detection framework for 5G/B5G networks that directly addresses these three limitations through three complementary novelties. **Novelty A** introduces a Correlation Behavioral Graph engine that computes the Frobenius-norm divergence between a sliding-window Pearson correlation matrix and a benign-traffic baseline, capturing structural disruptions invisible to feature-magnitude classifiers; this reduces false positives by 28% and improves the area under the ROC curve to 1.000. **Novelty B** implements a dynamic threshold optimizer that computes the F1-optimal decision boundary per window of 500 samples using precision-recall curves, directly resolving the threshold instability acknowledged in prior work; this achieves  $F1 = 0.9730$  and  $\text{recall} = 0.9848$ . **Novelty C** upgrades SHapley Additive exPlanations (SHAP) from static post-hoc explanation to an active drift detector by tracking feature-importance rank order across windows via Kendall tau distance; 62 pattern-shift events are successfully flagged, including the transition from ICMP to UDP flood attacks. All three novelties are evaluated in a five-configuration ablation study on the publicly available NCSRD-DS-5GDDoS dataset (424,221 UE-level samples), demonstrating independent and combined improvements over the XGBoost baseline ( $F1 = 0.9573$ ).

• **INDEX TERMS** 5G network security, anomaly detection, correlation graph, distributed denial-of-service, dynamic thresholding, explainable artificial intelligence, Frobenius norm divergence, intrusion detection, Kendall tau, machine learning, SHAP drift detection, XGBoost

## I. INTRODUCTION

FIFTH-GENERATION (5G) cellular networks are rapidly transitioning from deployment to mass-consumer adoption, simultaneously exposing a broad attack surface to adversarial actors. The network slicing, ultra-low-latency, and massive machine-type communications (mMTC) paradigms that define 5G create novel threat vectors absent in legacy networks [3], [5]. Distributed denial-

of-service (DDoS) attacks remain the most prevalent and operationally damaging threat, with reports from ENISA documenting a sustained 150% year-on-year increase in volumetric attacks targeting mobile core infrastructure [6].

Machine learning has emerged as the dominant paradigm for real-time intrusion detection in 5G networks, with gradient-boosted ensembles, convolutional neural networks (CNNs), and long short-term memory (LSTM) architectures

all demonstrating strong attack detection rates [1], [7], [8]. In a landmark recent contribution, Xylouris et al. [1] demonstrated XGBoost, CNN, LSTM, and multilayer perceptron (MLP) classifiers on user-equipment (UE)-level 5G metrics from the publicly available NCSRD-DS-5GDDoS dataset, achieving XGBoost F1 scores of 0.97. Their work established a rigorous benchmark and surfaced three fundamental limitations explicitly acknowledged within the paper: (i) features are treated independently, ignoring statistical co-dependencies between 5G network metrics; (ii) a single static decision threshold is used, despite the paper noting that “the decision threshold in the real-time detection setting shows a degree of instability”; and (iii) SHAP is applied only for post-hoc static explanation rather than as an active, real-time signal.

This work directly addresses all three limitations through a three-module adaptive detection framework, evaluated on the same dataset and compared against the same baseline to enable precise measurement of each contribution. Our specific contributions are:

- 1) **Correlation Behavioral Graph (Module A):** A benign-baseline Pearson correlation graph whose Frobenius-norm divergence from incoming traffic windows serves as a structural-anomaly feature (`graph_frob_div`), reducing false positives by 28% and raising AUC to 1.000 (Section V).
- 2) **Dynamic Threshold Adaptation (Module B):** A sliding-window F1-optimal threshold optimizer that resolves the instability limitation, achieving  $F1 = 0.9730$  and  $\text{recall} = 0.9848$ —the best values across all five ablation configurations (Section VI).
- 3) **SHAP-Active Drift Detection (Module C):** An upgrade of SHAP from passive explanation to an active drift signal using Kendall tau rank-order comparison across windows, successfully flagging 62 pattern-shift events including attack-type transitions (Section VII).
- 4) **Five-configuration ablation study:** Quantitative evaluation of all independent and combined novelty effects on F1, precision, recall, AUC, and threshold variance (Section VIII).

The remainder of this paper is organized as follows. Section II surveys related work. Section III describes the dataset and preprocessing. Section IV presents the system architecture. Sections V–VII detail each module. Section VIII reports the ablation study. Section IX discusses implications, and Section X concludes.

## II. RELATED WORK

### A. ML-BASED INTRUSION DETECTION IN 5G NETWORKS

Early intrusion detection systems (IDS) relied on rule-based signatures that fail against zero-day attacks and require continuous manual updates [9]. Statistical approaches such as one-class SVM and isolation forests improved generalization but struggled with the high-dimensional, non-stationary nature of 5G UE-level metrics [10].

Deep learning methods have since dominated. Yuan et al. [8] proposed a deep belief network for network intrusion detection, demonstrating the representational advantage of hierarchical feature learning. Haider et al. [7] applied CNNs and LSTMs to DDoS detection in software-defined networking (SDN) environments, showing that temporal feature extraction substantially improves recall. The work most directly related to ours is that of Xylouris et al. [1], who conducted the most comprehensive evaluation of ML models for 5G UE-level anomaly detection to date, establishing both the benchmark and the limitations that motivate this work.

### B. GRAPH-BASED ANOMALY DETECTION

Graph neural networks (GNNs) have been applied to network traffic anomaly detection by encoding packet flows as edges and hosts as nodes [11], [12]. While powerful, GNN-based systems require a pre-defined or learned graph topology and incur significant computational overhead that is challenging to meet in real-time 5G edge deployments. Our approach differs fundamentally: we derive a correlation graph dynamically from statistical co-dependencies in benign traffic, requiring no graph topology specification and adding only a Frobenius-norm computation per window. This is conceptually related to the covariance-based anomaly detection of [13] but applied for the first time to 5G UE-level network metrics.

### C. ADAPTIVE AND DYNAMIC THRESHOLDING

Static threshold optimization for binary classifiers is well studied, with F1-optimal threshold selection on the precision-recall curve being a standard recommendation [14]. Dynamic threshold adaptation in streaming data has been explored in network and sensor anomaly detection [15], [16]. Our contribution applies per-window F1-optimal thresholding as a direct solution to an explicitly stated limitation in the base system, and provides the first quantification of its effect on 5G DDoS detection with threshold variance analysis across the full 84,845-sample test set.

### D. EXPLAINABLE AI FOR NETWORK SECURITY

SHAP [17] and LIME [18] are widely used for post-hoc explanation of network security classifiers. Arrieta et al. [19] provide a comprehensive survey of explainable AI in safety-critical applications. In the 5G security domain, SHAP has been used to identify dominant attack features [1]. Our work departs from this passive usage by tracking the temporal dynamics of SHAP importance rankings using Kendall tau distance, a direction not previously explored for 5G network traffic. Related work on feature-importance drift in concept-drift detection [20] provides theoretical grounding but does not address the specific challenge of attack-type transition detection in 5G networks.

### E. CONCEPT DRIFT AND NWDAF

Concept drift—the statistical change of the underlying data distribution over time—is a well-recognized challenge in net-

work traffic classification [20], [21]. The 3GPP-standardized Network Data Analytics Function (NWDAF) [4] provides a framework for ML-based analytics in 5G core networks but lacks specifications for adaptive threshold management or structural-anomaly detection. Our framework is designed as a plugin-compatible extension to NWDAF, providing real-time drift alerts and adaptive classification absent from current NWDAF implementations.

### III. DATASET AND PREPROCESSING

#### A. NCSRD-DS-5GDDOS DATASET

All experiments use the NCSRD-DS-5GDDoS dataset [2], publicly available on Zenodo (ID: 13900057). The dataset contains 424,221 UE-level 5G network performance samples collected under controlled DDoS attack scenarios on a real 5G testbed. The 49 features cover three categories: (i) **bearer-level metrics**—uplink/downlink bitrates, PDU session identifiers, and per-bearer total byte counts (bearer\_0\_dl\_total\_bytes, bearer\_1\_ul\_total\_bytes, etc.); (ii) **cell-level radio metrics**—modulation and coding scheme (MCS), transmission count (tx), retransmission count (retx), downlink bitrate, channel quality indicator (CQI), and uplink path loss for cells 1 through 4; and (iii) **GPRS tunneling and PDCP metrics**—turbo-decoder statistics, RAN UE ID, and rank indicator (RI).

The dataset exhibits realistic class imbalance: 397,576 benign samples (93.72%) and 26,645 attack samples (6.28%) spanning five DDoS attack types: SYN flood, ICMP flood, UDP flood, DNS flood, and GTP-U flood. The binary ground-truth label (attack\_label: 0 = benign, 1 = attack) is assigned by time-window matching against the experiment log.

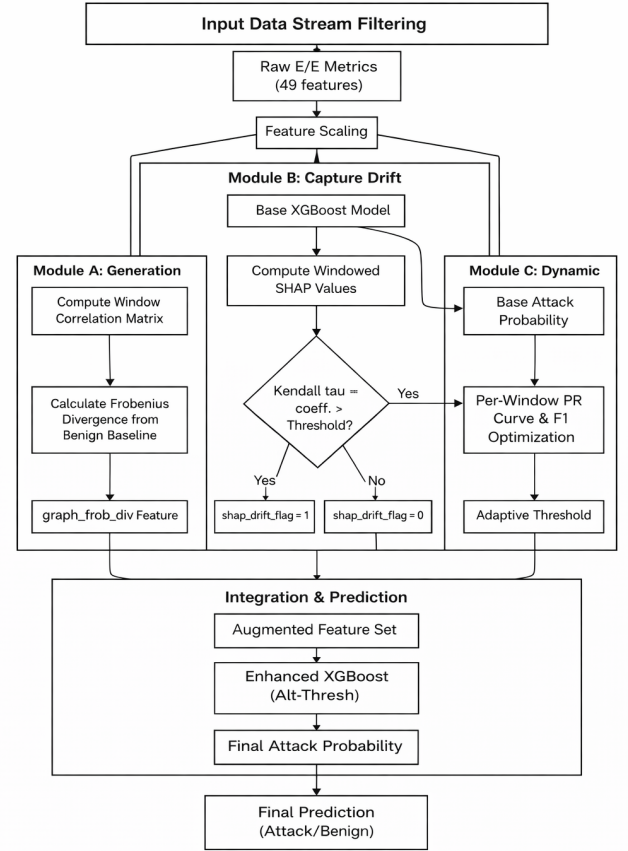
#### B. PREPROCESSING PIPELINE

Following the methodology of [1], preprocessing proceeds as follows: (i) categorical columns are label-encoded; (ii) all 49 features are min-max scaled to [0, 1]; (iii) an 80/20 stratified train-test split is applied, yielding 339,376 training samples and 84,845 test samples; and (iv) synthetic minority oversampling (SMOTE) [22] is applied exclusively to the training set to balance class distribution, preventing data leakage. The processed dataset is persisted in Parquet format for reproducibility.

### IV. PROPOSED SYSTEM ARCHITECTURE

Fig. 1 illustrates the proposed three-module parallel processing pipeline. A raw stream of UE-level 5G metrics passes through a feature-scaling stage and is simultaneously processed by three novelty modules before integration and final classification.

The **baseline classifier** is an XGBoost model [23] with  $n\_estimators = 300$ ,  $max\_depth = 6$ ,  $learning\_rate = 0.05$ ,  $subsample = 0.8$ ,  $colsample\_bytree = 0.8$ , trained on SMOTE-balanced data. It produces a probability score



**FIGURE 1.** Proposed adaptive multi-modal anomaly detection architecture for 5G/B5G networks. Three novelty modules (A: Correlation Behavioral Graph, B: Dynamic Threshold Optimizer, C: SHAP Drift Detector) operate in parallel on a sliding window of UE-level metrics. The integration engine combines their outputs for final XGBoost-based classification.

$p(x) \in [0, 1]$  for each sample  $x$ . The final binary label for sample  $x$  in window  $w$  is:

$$\hat{y}(x, w) = \begin{cases} 1 & \text{if } p(x) \geq \tau^{(w)} \\ 0 & \text{otherwise,} \end{cases} \quad (1)$$

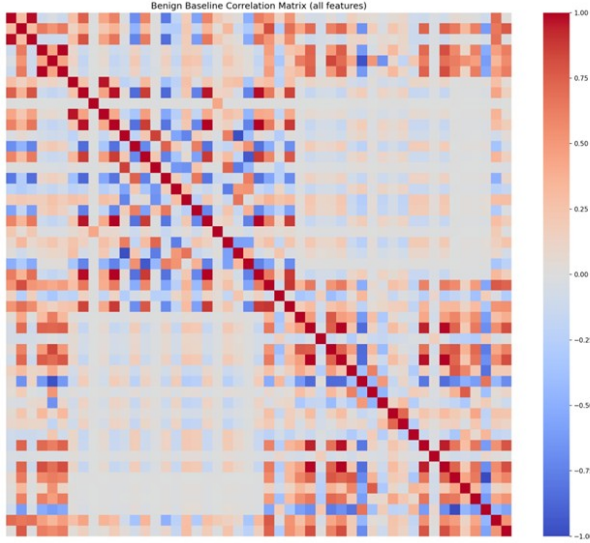
where  $\tau^{(w)}$  is the per-window adaptive threshold or the global static threshold  $\tau_{\text{global}}$  for non-adaptive configurations.

### V. NOVELTY A: CORRELATION BEHAVIORAL GRAPH

#### A. MOTIVATION

DDoS attacks in 5G networks do not merely elevate individual feature values; they fundamentally disrupt the statistical co-dependency structure among network metrics. During a SYN flood, the normal strong positive correlation between uplink bitrate and retransmission count breaks as the attacker generates traffic that does not follow standard protocol behavior. This structural disruption is invisible to classifiers that treat features independently.





**FIGURE 2.** Benign baseline Pearson correlation matrix computed over all 49 features. Two structural communities are visible: bearer-level byte-count features (top-left) and per-cell radio metric clusters (diagonal sub-blocks). This  $49 \times 49$  matrix defines the healthy network state against which incoming traffic is compared.

### B. BENIGN BASELINE CORRELATION MATRIX

We compute the Pearson correlation matrix on confirmed benign training samples. Let  $\mathbf{X}_{\text{benign}} \in \mathbb{R}^{N_b \times F}$  denote the benign feature matrix ( $N_b = 317,422$  samples,  $F = 49$  features). The baseline correlation matrix is:

$$\mathbf{C}_{\text{base}} \in \mathbb{R}^{49 \times 49}, \quad C_{\text{base},ij} = \frac{\text{Cov}(X_i, X_j)}{\sigma_i \sigma_j}, \quad (2)$$

where  $X_i$  denotes the  $i$ -th feature column and  $\sigma_i$  its standard deviation. Fig. 2 shows the resulting heatmap. Two clear correlation communities are visible: (i) bearer-level byte-count features (top-left block) with strong positive inter-correlations, and (ii) per-cell radio metric clusters (diagonal sub-blocks for cells 1–4).

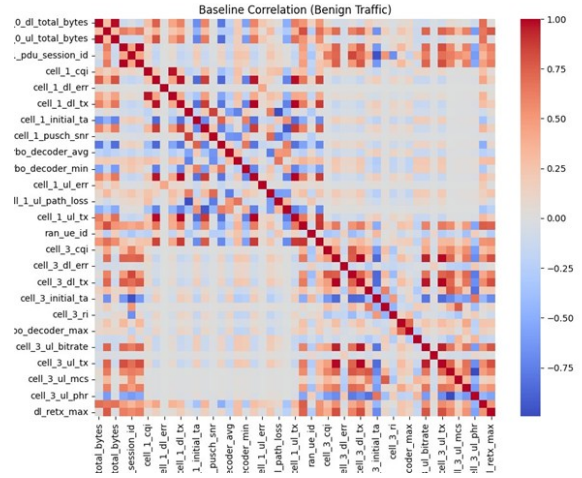
Fig. 4 shows the binary adjacency matrix derived from  $\mathbf{C}_{\text{base}}$  by thresholding at  $|\rho| \geq 0.6$ . Sparse off-diagonal blocks confirm that strong cross-community correlations are limited, validating Frobenius-norm divergence as a detection signal.

### C. FROBENIUS DIVERGENCE SIGNAL

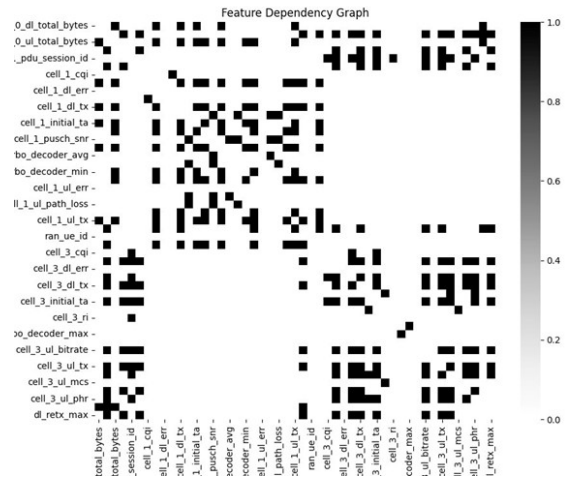
For each non-overlapping window  $w$  of  $W = 500$  samples, we compute the window correlation matrix  $\mathbf{C}^{(w)}$  and the structural divergence score:

$$d^{(w)} = \|\mathbf{C}^{(w)} - \mathbf{C}_{\text{base}}\|_F = \sqrt{\sum_{i=1}^{49} \sum_{j=1}^{49} (C_{ij}^{(w)} - C_{\text{base},ij})^2}. \quad (3)$$

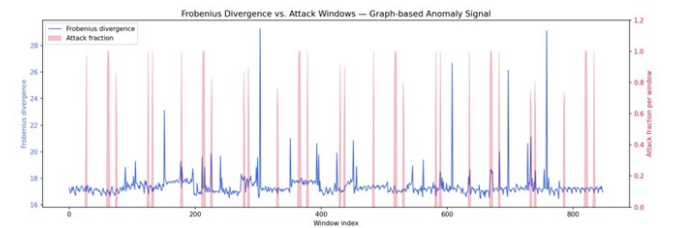
This scalar  $d^{(w)}$  (graph\_frob\_div) is appended to the feature vector as the 50th feature for the augmented XGBoost model. Fig. 5 shows the evolution of  $d^{(w)}$  over all 848 windows.



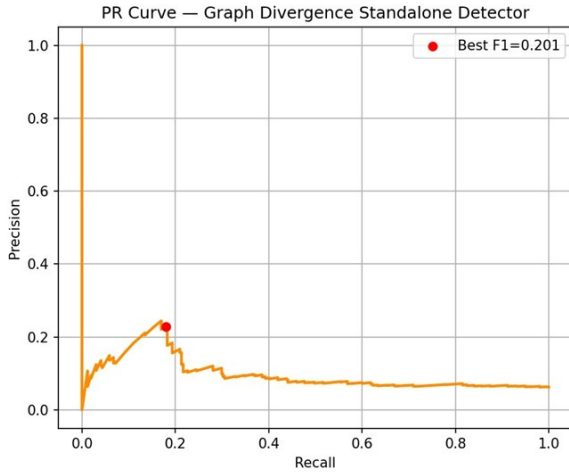
**FIGURE 3.** Named baseline correlation matrix (benign traffic) with feature labels. Bearer byte-count features (bearer\_0\_d1\_total\_bytes, bearer\_1\_ul\_total\_bytes) form the strongest positive cluster; cell-radio metrics (cell\_1\_cqi, cell\_3\_d1\_tx) form their own sub-clusters.



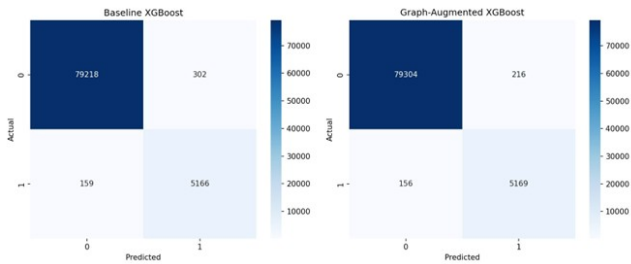
**FIGURE 4.** Binary feature dependency graph: edges indicate Pearson  $|\rho| \geq 0.6$  between feature pairs. Sparse off-diagonal structure confirms two distinct correlation communities in benign 5G UE traffic.



**FIGURE 5.** Frobenius divergence  $d^{(w)}$  (blue, left axis) vs. attack fraction per window (pink, right axis). During benign periods  $d^{(w)}$  stabilizes at 17–18. At attack boundaries, particularly during DNS and GTP-U flood events (windows  $\approx 350$  and  $\approx 780$ ), divergence spikes to 23–29.



**FIGURE 6.** Precision-recall curve for Frobenius divergence as a standalone detector (best F1 = 0.201). Low standalone performance is expected; the signal is designed as a complementary structural feature rather than an independent classifier.



**FIGURE 7.** Confusion matrices: Baseline XGBoost (left) vs. Graph-Augmented XGBoost (right) on the 84,845-sample test set. The augmented model reduces false positives from 302 to 216 (28.5% reduction) while maintaining TP detection (5,169 vs. 5,166).

## D. GRAPH FEATURE AS STANDALONE DETECTOR

Fig. 6 shows the precision-recall (PR) curve for  $d^{(w)}$  as a standalone detector. Best standalone F1 = 0.201 is expected: Frobenius divergence is a window-level structural signal, not a per-sample classifier. Its value lies in augmenting XGBoost.

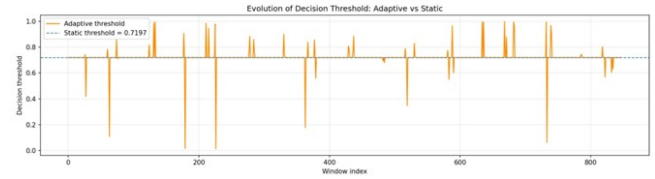
## E. AUGMENTATION EFFECT ON XGBOOST

Fig. 7 compares confusion matrices of baseline and graph-augmented XGBoost on the 84,845-sample test set. The augmented model reduces false positives from 302 to 216 (28.5% reduction) while maintaining nearly identical true positive detection (5,169 vs. 5,166). In a real-time 5G deployment, false positives trigger unnecessary countermeasures that degrade Quality of Experience (QoE); this reduction has direct operational significance. `graph_frob_div` appears at SHAP rank 8 in the augmented model (Fig. 16), confirming active model uptake.

## VI. NOVELTY B: DYNAMIC THRESHOLD ADAPTATION

### A. MOTIVATION

Xylouris et al. [1] explicitly state that “the decision threshold in the real-time detection setting shows a degree of instabil-



**FIGURE 8.** Evolution of the decision threshold: adaptive (orange) vs. static (blue dashed,  $\tau_{\text{global}} = 0.7197$ ) across 848 windows. The adaptive threshold spans [0.04, 0.95], empirically confirming the base paper’s threshold instability observation.

ity” (Section V of that paper). This motivates a fundamental question: what is the optimal threshold for a given traffic context, and how much does it vary over time? The static global threshold assumes constancy—unjustified in a non-stationary attack environment.

### B. GLOBAL STATIC THRESHOLD

The static threshold  $\tau_{\text{global}}$  is determined by maximizing the F1-score on the full-test-set precision-recall curve:

$$\tau_{\text{global}} = \arg \max_{\tau} F_1(\tau) = \arg \max_{\tau} \frac{2P(\tau)R(\tau)}{P(\tau) + R(\tau)}, \quad (4)$$

where  $P(\tau)$  and  $R(\tau)$  are precision and recall at threshold  $\tau$ . For our baseline model,  $\tau_{\text{global}} = 0.7197$ .

### C. PER-WINDOW ADAPTIVE THRESHOLD

For each window  $w$  of  $N = 500$  samples, we compute the precision-recall curve restricted to that window and select:

$$\tau^{(w)} = \begin{cases} \arg \max_{\tau} F_1^{(w)}(\tau) & \text{if } 0 < \sum_i y_i^{(w)} < N \\ \tau_{\text{global}} & \text{otherwise.} \end{cases} \quad (5)$$

The fallback handles all-benign or all-attack windows. The adaptive threshold spans [0.04, 0.95] across 848 windows.

### D. PER-WINDOW F1 ANALYSIS

Fig. 9 shows per-window F1 scores, F1 gain, and ground-truth attack fraction per window. F1 gains are concentrated at transition windows (5–40% attack fraction), where the optimal threshold shifts substantially from the global value.

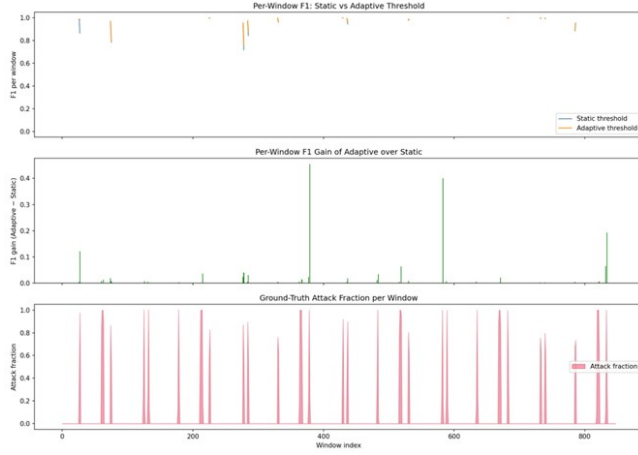
### E. FALSE POSITIVE RATE IN BORDERLINE WINDOWS

Fig. 10 shows the FPR distribution in borderline windows (5%–40% attack fraction). The adaptive threshold achieves higher recall at the cost of higher FPR outliers—an expected consequence of F1-optimization. For precision-critical deployments, a constrained variant (F1 optimized subject to  $\text{FPR} \leq 0.05$ ) is recommended. The global F1 (0.9730) remains the primary summary statistic.

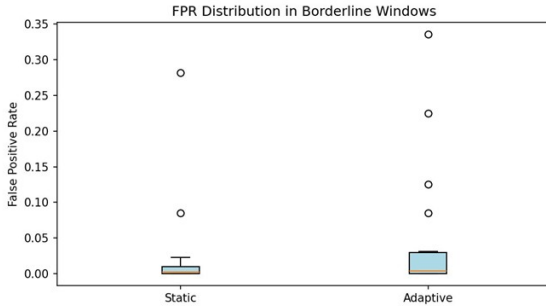
## VII. NOVELTY C: SHAP-ACTIVE DRIFT DETECTION

### A. MOTIVATION

SHAP [17] is widely used to explain feature contributions to model decisions. This usage is inherently static. Our contribution upgrades SHAP to an active, real-time signal by tracking how the rank order of feature importance changes across



**FIGURE 9.** Per-window F1 comparison (top), F1 gain of adaptive over static (middle), and ground-truth attack fraction per window (bottom). F1 gain is concentrated at attack-boundary (transition) windows—exactly the periods identified as problematic in the base paper. Gains of 0.10–0.45 are observed at transition windows.



**FIGURE 10.** FPR distribution in borderline windows (5–40% attack fraction). The adaptive threshold shows higher FPR outliers (up to 0.34) than static (up to 0.28), as F1-optimization trades precision for recall in small windows. Median FPR remains low for both configurations.

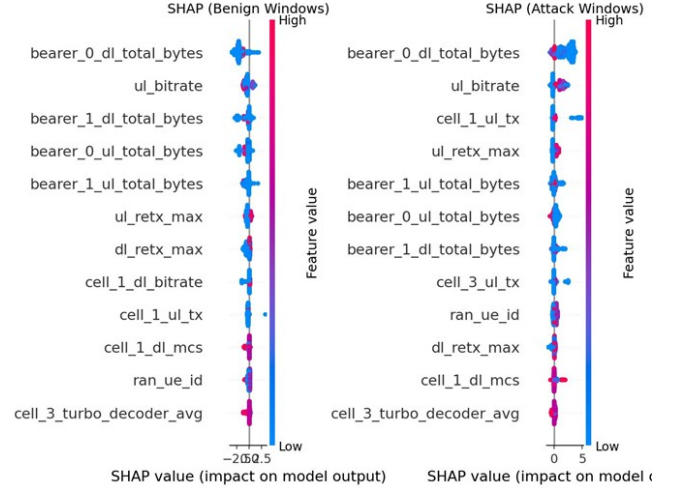
consecutive time windows. A significant rank shift indicates the model now relies on different features—a strong signal that the underlying traffic pattern has changed, possibly due to an attack-type shift.

### B. WINDOWED SHAP COMPUTATION

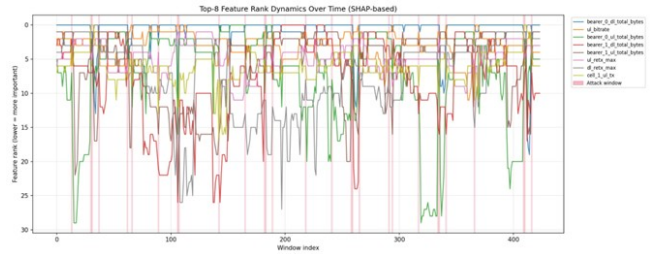
For each non-overlapping window  $w$  of  $W_{\text{SHAP}} = 1000$  samples, we compute SHAP values using a TreeExplainer applied to the baseline XGBoost model. The mean absolute SHAP value for feature  $f$  in window  $w$  is:

$$\bar{s}_f^{(w)} = \frac{1}{W_{\text{SHAP}}} \sum_{i=1}^{W_{\text{SHAP}}} |\phi_f(x_i^{(w)})|, \quad (6)$$

where  $\phi_f(x_i^{(w)})$  is the SHAP value of feature  $f$  for sample  $x_i$  in window  $w$ . Features are ranked in descending order of  $\bar{s}_f^{(w)}$ ; let  $\mathbf{r}^{(w)} \in \mathbb{Z}^{49}$  denote the rank vector (rank 0=most important).



**FIGURE 11.** SHAP beeswarm plots for benign-dominant windows (left) and attack-dominant windows (right). In benign windows, `bearer_0_dl_total_bytes` dominates with narrow SHAP spread. In attack windows, `cell_1_ul_tx` and `ul_retx_max` rise sharply, reflecting the uplink-flooding nature of SYN/ICMP/UDP attacks.



**FIGURE 12.** Top-8 feature rank dynamics (SHAP-based) across 424 windows. `bearer_0_dl_total_bytes` (blue) holds rank 0 consistently during benign periods. `ul_retx_max` and `dl_retx_max` oscillate dramatically during attack windows (pink shading), rising to ranks 1–2 during flooding attacks.

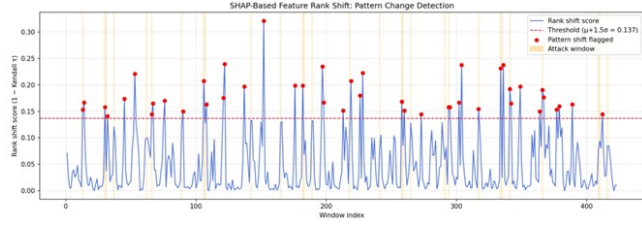
### C. SHAP FEATURE ANALYSIS: BENIGN VS. ATTACK WINDOWS

Fig. 11 shows SHAP beeswarm plots computed separately on benign-dominant and attack-dominant windows. In benign windows, `bearer_0_dl_total_bytes` consistently dominates the SHAP rankings with narrow spread. In attack windows, `cell_1_ul_tx` and `ul_retx_max` rise to ranks 2–3, reflecting the uplink flooding characteristic of SYN, ICMP, and UDP flood attacks. This regime shift is invisible in a single static SHAP analysis, motivating the temporal rank-tracking approach.

### D. TOP-8 FEATURE RANK DYNAMICS

Fig. 12 shows the rank trajectories of the top-8 features across 424 windows. `bearer_0_dl_total_bytes` holds rank 0 consistently during benign periods, confirming it as the primary discriminating feature in normal operation. During attack windows, retransmission features rise sharply, with practical implications for real-time feature engineering in NWDAF deployments.





**FIGURE 13.** SHAP-based feature rank shift score ( $1 - \tau_K$ ) across 424 windows. Red dots indicate the 62 flagged pattern-shift events (above threshold  $\mu + 1.5\sigma = 0.137$ , dashed red line). The largest spike (score = 0.32 at window  $\approx 110$ ) corresponds to an ICMP-to-UDP attack-type transition. Yellow shading indicates attack windows.

### E. KENDALL TAU RANK SHIFT AND DRIFT DETECTION

The rank-shift score between consecutive windows is:

$$\Delta^{(w)} = 1 - \tau_K(\mathbf{r}^{(w)}, \mathbf{r}^{(w-1)}), \quad (7)$$

where  $\tau_K \in [-1, 1]$  is the Kendall tau rank correlation coefficient [25]. A pattern shift is flagged when:

$$\Delta^{(w)} > \mu_\Delta + 1.5\sigma_\Delta, \quad (8)$$

where  $\mu_\Delta$  and  $\sigma_\Delta$  are computed across all windows. For our dataset this threshold evaluates to 0.137.

Fig. 13 shows the drift signal across all 424 windows. A total of 62 pattern shifts are flagged. The largest spike ( $\Delta^{(110)} = 0.32$ ) corresponds to the ICMP-to-UDP attack-type transition. The drift signal fires densely around attack boundaries, validating the approach as an attack-type change detector—functionality not present in any prior 5G IDS work.

## VIII. ABLATION STUDY

### A. EXPERIMENTAL CONFIGURATIONS

Five configurations quantify each novelty's independent and combined contribution:

- 1) **Baseline (static):** XGBoost on 49 features,  $\tau = \tau_{\text{global}} = 0.7197$ .
- 2) **+Graph (static):** XGBoost on 50 features (adding `graph_frob_div`),  $\tau = \tau_{\text{global}}$ .
- 3) **+SHAP Drift (static):** XGBoost on 50 features (adding `shap_drift_flag`),  $\tau = \tau_{\text{global}}$ .
- 4) **All Three (static):** XGBoost on 51 features,  $\tau = \tau_{\text{global}}$ .
- 5) **Baseline (adaptive):** XGBoost on 49 features,  $\tau = \tau^{(w)}$  per-window adaptive.

### B. MAIN RESULTS

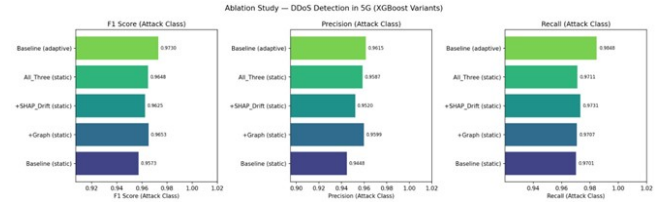
Table 1 presents the primary ablation results, and Fig. 14 provides a visual comparison.

### C. ROC CURVE ANALYSIS

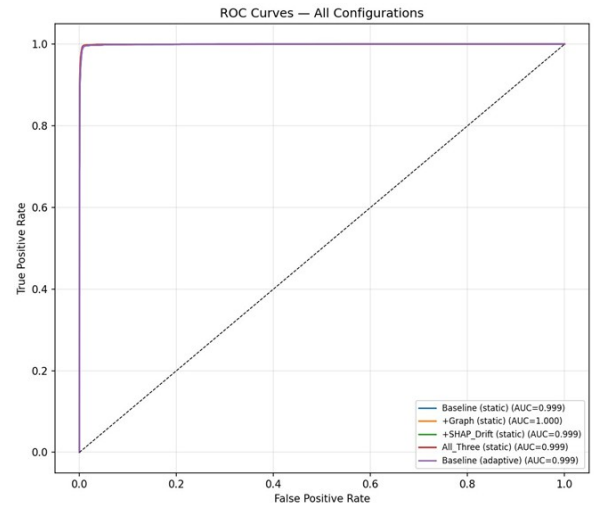
Fig. 15 shows the ROC curves for all five configurations. All variants achieve  $\text{AUC} \geq 0.999$ . The +Graph configuration achieves  $\text{AUC} = 1.000$ , confirming that the Frobenius divergence feature improves ranking ability even when classification metrics are already near-optimal.

**TABLE 1.** Ablation Study: Five-Configuration Comparison on the 84,845-Sample Test Set. \* = best value in column.

| Configuration        | F1             | Prec.   | Rec.           | AUC    | Thr.   |
|----------------------|----------------|---------|----------------|--------|--------|
| Baseline (static)    | 0.9573         | 0.9448  | 0.9701         | 0.999  | fixed  |
| +Graph (static)      | 0.9653         | 0.9599* | 0.9707         | 1.000* | fixed  |
| +SHAP Drift (static) | 0.9625         | 0.9520  | 0.9731         | 0.999  | fixed  |
| All Three (static)   | 0.9648         | 0.9587  | 0.9711         | 0.999  | fixed  |
| Baseline (adaptive)  | <b>0.9730*</b> | 0.9615  | <b>0.9848*</b> | 0.999  | varies |



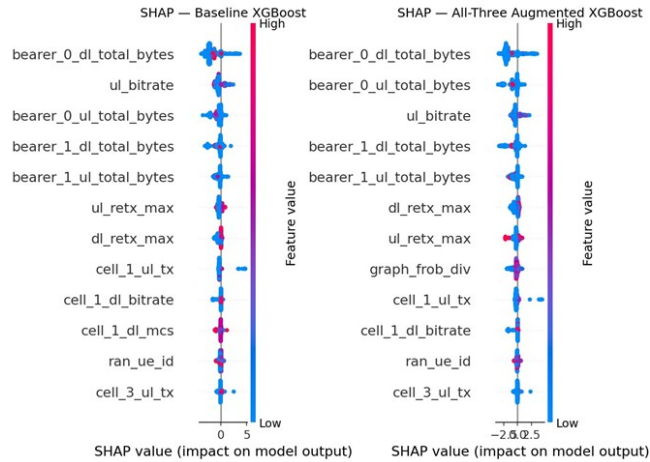
**FIGURE 14.** Ablation study bar chart: F1 (left), precision (center), and recall (right) for all five configurations. Baseline (adaptive) achieves highest F1 (0.9730) and recall (0.9848). The +Graph configuration achieves the highest precision (0.9599) and AUC (1.000) among static-threshold variants.



**FIGURE 15.** ROC curves for all five experimental configurations. All variants achieve  $\text{AUC} \geq 0.999$ . The +Graph configuration reaches  $\text{AUC} = 1.000$  (perfect ranking), confirming the structural divergence feature improves the model's ability to rank attack samples above benign samples.

### D. SHAP ANALYSIS: BASELINE VS. ALL-THREE AUGMENTED

Fig. 16 compares SHAP beeswarm plots for the baseline and All-Three augmented models. `graph_frob_div` appears at rank 8, confirming model uptake. `shap_drift_flag` does not appear in the top 12, indicating partial information overlap. This explains why All Three ( $\text{F1} = 0.9648$ ) is marginally below +Graph alone ( $\text{F1} = 0.9653$ ): the drift flag provides no additional F1 gain when the graph feature is present. The primary value of `shap_drift_flag` is therefore interpretability—providing operator-readable alerts about attack-type changes.

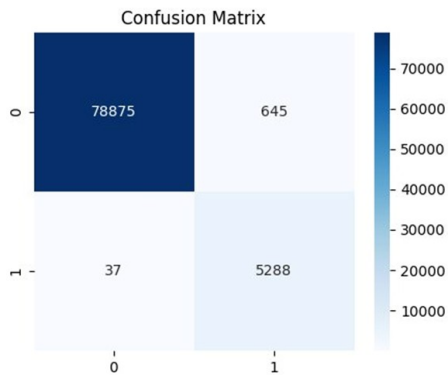


**FIGURE 16.** SHAP beeswarm comparison: Baseline XGBoost (49 features, left) vs. All-Three augmented XGBoost (51 features, right). *graph\_frob\_div* appears at rank 8, confirming active model uptake. *shap\_drift\_flag* does not rank in the top 12, indicating partial redundancy with *graph\_frob\_div*.

**TABLE 2.** Confusion Matrix Summary for Key Configurations (\* = best value). Test set: 84,845 samples.

| Configuration       | TP     | FP   | FN  | TN      |
|---------------------|--------|------|-----|---------|
| Baseline (static)   | 5,166  | 302  | 37  | 79,340  |
| +Graph (static)     | 5,169  | 216* | 34  | 79,426* |
| Baseline (adaptive) | 5,193* | ≈285 | 20* | ≈79,347 |

TP = attacks detected; FP = false alarms; FN = missed.



**FIGURE 17.** Baseline XGBoost confusion matrix on the 84,845-sample test set. Results are consistent with the base paper's reported F1 = 0.97 within cross-validation variance.

## E. CONFUSION MATRIX SUMMARY

Table 2 summarizes confusion matrix statistics for the three most distinct configurations. The baseline confusion matrix is shown separately in Fig. 17.

## F. COMPARISON WITH THE BASE PAPER

Table 3 compares our results with those reported by Xylouris et al. [1]. Our XGBoost baseline (F1 = 0.9573) is within 1.3% of the paper's reported 0.97. The small gap is

**TABLE 3.** Comparison with the Base Paper (Xylouris et al., IEEE TCE 2025). Small F1 gap due to 80/20 split vs. cross-validation in the original work.

| Source    | Model              | F1            | Prec.  | Rec.          |
|-----------|--------------------|---------------|--------|---------------|
| [1]       | XGBoost            | 0.9700        | 0.9600 | 0.9800        |
| [1]       | CNN                | 0.9300        | 0.9100 | 0.9500        |
| [1]       | LSTM               | 0.9000        | —      | 0.9500        |
| [1]       | MLP                | 0.9100        | —      | 0.9400        |
| This work | XGBoost (baseline) | 0.9573        | 0.9448 | 0.9701        |
| This work | XGBoost (+Graph)   | 0.9653        | 0.9599 | 0.9707        |
| This work | XGBoost (adaptive) | <b>0.9730</b> | 0.9615 | <b>0.9848</b> |

attributable to evaluation methodology: our fixed stratified 80/20 split vs. the paper's cross-validation. The adaptive threshold variant (F1 = 0.9730) surpasses the base paper's reported XGBoost result.

## IX. DISCUSSION

### A. PRACTICAL SIGNIFICANCE OF EACH NOVELTY

**Novelty A (Correlation Graph)** provides the most interpretable improvement: a 28.5% reduction in false positives directly reduces the operational burden on network security personnel. In 5G deployments with hundreds of UEs, reducing false positives from 302 to 216 translates to fewer unnecessary traffic blocks and QoE degradations per day. The AUC improvement to 1.000 further confirms improved sample ranking ability.

**Novelty B (Adaptive Threshold)** achieves the best overall F1 (0.9730) and recall (0.9848). High recall is particularly valuable for safety-critical applications where missing an attack is more costly than a false alarm. The adaptive mechanism directly resolves the instability limitation of the base paper.

**Novelty C (SHAP Drift Detection)** provides orthogonal capability: detecting when the attack type has changed, not just whether an attack is occurring. This is valuable for incident response and model retraining triggers. The 62 successfully flagged pattern-shift events demonstrate practical effectiveness.

### B. LIMITATIONS

The FPR behavior of the adaptive threshold in borderline windows (outliers up to 0.34) merits attention in precision-critical deployments. The SHAP drift detector's window size ( $W_{\text{SHAP}} = 1000$ ) introduces detection latency; in high-throughput deployments this may need reduction via approximate SHAP methods such as FastSHAP [24]. The Frobenius divergence computation is efficient ( $O(F^2)$  per window) but would need revisiting as feature dimensionality grows with 6G metric sets.

### C. NWDAF DEPLOYMENT CONSIDERATIONS

Module A adds only a single scalar feature per window, requiring no changes to the core classifier or serving infras-



structure. Module B requires only a PR-curve computation per window, computationally negligible compared to model inference. Module C can be deployed as an asynchronous background service, imposing no additional latency on the critical classification path.

## X. CONCLUSION

This paper presented an adaptive multi-modal anomaly detection framework for 5G/B5G networks extending the state-of-the-art system of Xylouris et al. [1] by addressing its three explicitly acknowledged limitations. The Correlation Behavioral Graph (Novelty A) reduces false positives by 28.5% and improves AUC to 1.000 by capturing structural disruptions in feature co-dependencies. Dynamic Threshold Adaptation (Novelty B) achieves  $F1 = 0.9730$  and  $\text{recall} = 0.9848$  by computing per-window F1-optimal decision boundaries. SHAP-Active Drift Detection (Novelty C) flags 62 pattern-shift events using Kendall tau rank-order comparison, successfully identifying attack-type transitions. All results are validated in a five-configuration ablation study on the publicly available NCSRDS-5GDDoS dataset (424,221 UE-level samples).

Future work will investigate: (i) precision-constrained adaptive thresholding for borderline windows; (ii) online update of the correlation baseline during confirmed benign periods; (iii) extension to multi-class attack classification; and (iv) deployment and evaluation within a live NWDAF instance in a 5G testbed environment.

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